

MINI-PROJECT SHORT TECHNICAL REPORT

Course: Cross-Platform Mobile App Development (VKU)
Mini-Project Title: Mini-Project 1 — Week 3 PWA (Trovey Field Survey)
Team / Student Name: Nguyễn Minh Duy
Submission Date: 03/09/2026

1. GENERAL INFORMATION & DELIVERABLE LINKS

Team Members:

1. Nguyễn Minh Duy — Student ID: 23IT038 — Role: Solo (UI, PWA, sync, deploy) — Contribution: 100%

Live Demo URL: <https://app.puretrovey.net/>

GitHub Repository: <https://github.com/minhduy6868/MOB-Survey>

Video Demo: Not submitted.

2. FEATURE IMPLEMENTATION CHECKLIST

#	Required Feature	Status	Implementation Details & Acceptance Level
1	Responsive mobile viewport	Complete	Flutter web, Material 3 NavigationBar, 360–430 px. Light/dark in Settings.
2	Web App Manifest (installable)	Complete	app/web/manifest.json: display standalone, start_url/scope HTTPS, icons 192/512 + maskable.
3	Service Worker lifecycle	Complete	app/web/sw.js: install (precache), activate (drop old caches), fetch. Cache name trovey-shell-v9.
4	Five caching strategies	Complete	Cache-First shell; Network-First /survey-template.json; SWR /sw-stats.json; Cache-Only /offline; Network-Only /api/*.
5	Local offline persistence	Complete	Hive boxes tickets-v3, settings, meta. On web this is IndexedDB.
6	Offline draft + submit queue	Complete	Draft saved in Hive. Submit sets queued. Status: draft queued syncing synced failed.
7	Automatic background sync	Complete	connectivity_plus drain on reconnect. SW sync tag trovey-sync posts DRAIN_QUEUE to clients.
8	HTTPS live deploy	Complete	Cloudflare Pages. Canonical host https://app.puretrovey.net/ .

3. TECHNICAL ARCHITECTURE & PROJECT STRUCTURE

Client is Flutter web. TroveyStore (ChangeNotifier) is the only app state. It owns Hive boxes, connectivity, and the sync queue. Screens read the store and call put / enqueue / drainQueue.

Directory map:

app/lib/data/store.dart	Hive + queue (tickets-v3)
app/lib/data/sync.dart	POST /api/sync, GET /api/records
app/lib/screens/	Home, form (3 steps), results, settings
app/web/manifest.json	Install metadata
app/web/sw.js	SW install / activate / fetch + sync
functions/api/sync.js	Write Cloudflare KV (TROVEY_RECORDS)
functions/api/records.js	JSON / CSV read API
sheets/Code.gs	Sheet pulls KV; not the primary store

State flow: form edit -> Hive draft -> Submit -> status queued -> SyncClient.send -> KV put ticket:{clientId}. clientId is a UUID, so retry does not create a second row. On online, pullCloud() merges remote rows, then drainQueue() sends local queued/failed rows.

Exception handling: HTTP or body ok!=true / kv!=true throws. sendOne() sets status failed and lastError. pullCloud() swallows network errors and keeps the local box. The UI shows queued / failed / retry. A ticket is accepted when KV write succeeds. Sheet webhook errors do not fail the ticket.

Cache map used in the report demo: navigations try network then fall back to cached /index.html or /offline; API is never cached.

4. EMPIRICAL EVIDENCE & SCREENSHOTS

Screenshots from the live URL on a 390x844 viewport (Chromium). Captured 03/09/2026.



Figure 1. Home: topic, start CTA, submitted/queue counts, install card.

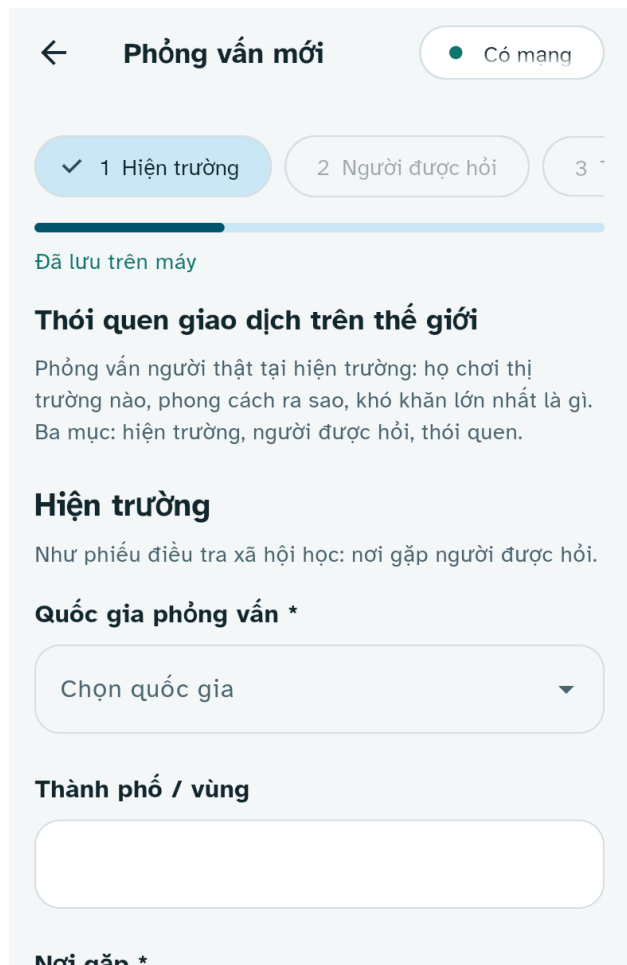
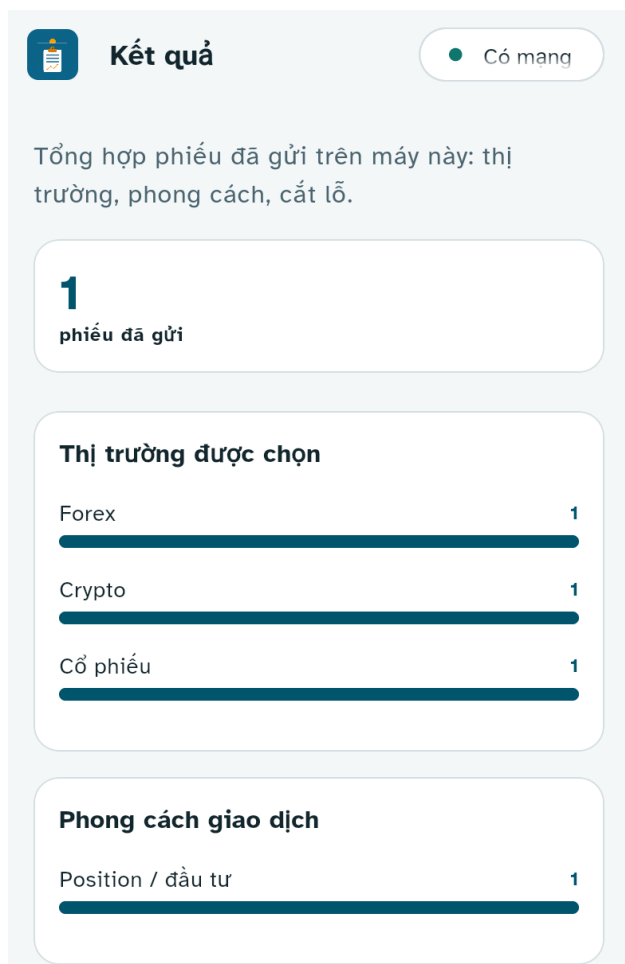


Figure 2. Form step 1 (site): country dropdown and required place field.



Kết quả Có mạng

Tổng hợp phiếu đã gửi trên máy này: thị trường, phong cách, cắt lỗ.

1
phiếu đã gửi

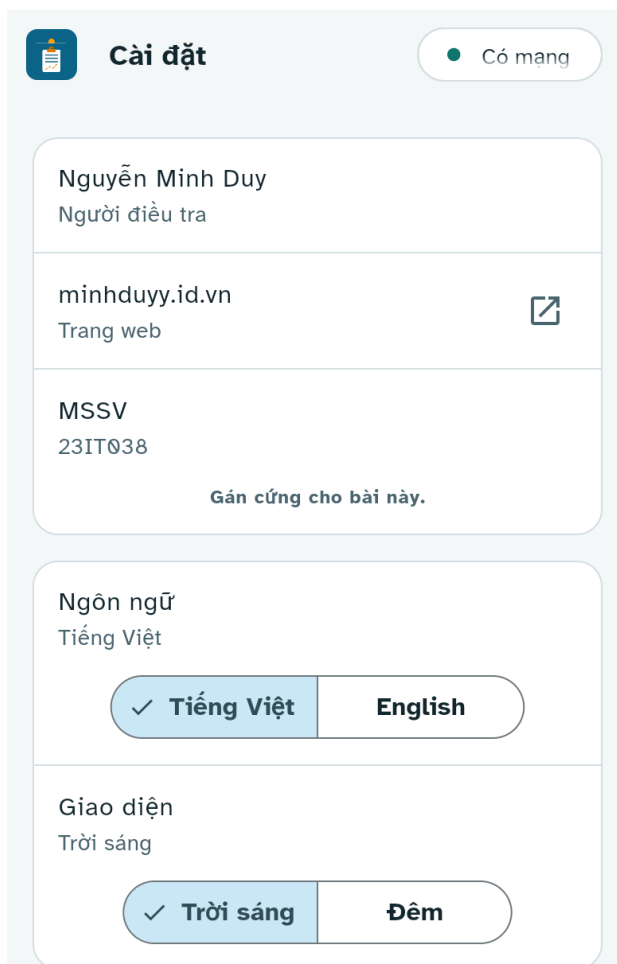
Thị trường được chọn

Forex	1
Crypto	1
Cổ phiếu	1

Phong cách giao dịch

Position / đầu tư	1
-------------------	---

Figure 3. Results: local totals after submit.



Cài đặt Có mạng

Nguyễn Minh Duy
Người điều tra

minhduy.id.vn
Trang web

MSSV
23IT038

Gán cứng cho bài này.

Ngôn ngữ
Tiếng Việt

✓ Tiếng Việt English

Giao diện
Trời sáng

✓ Trời sáng Đêm

Figure 4. Settings: collector 23IT038 (read-only), locale, theme.

Cache-Only fallback /offline is in the live host (Phiếu vẫn ở trên máy). It is used when a navigation miss is not in cache.

5. TECHNICAL CHALLENGES & RESOLUTIONS

5.1 Chrome ERR_FAILED on the apex host

<https://puretrovey.net/> returned 302 to the app host, but some campus Chrome sessions showed ERR_FAILED. The origin was up. The browser reused a cached Alt-Svc HTTP/3 (h3) route that did not complete on that network. Change: document <https://app.puretrovey.net/> as the demo URL; send Alt-Svc: clear on / and /index.html; service worker uses network-first for navigations. Workaround for an old home-screen icon: open the app host or <https://app.puretrovey.net/?reset=1>.

5.2 Google Sheets webhook returned 401

Apps Script doPost required a signed-in user (Who has access != Anyone), so the webhook could not be the database. Change: Cloudflare KV is the store. POST /api/sync writes KV and returns ok only if kv is true. The sheet runs beautifyAndSync() / UrlFetchApp on GET /api/records. Sheet downtime does not block submit.